

Bridging Structured Knowledge and Data: A Unified Framework with Finance Applications

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Introduction: The Data-Driven Paradigm

Machine learning models have achieved impressive performance by learning *complex, nonlinear* patterns directly from vast datasets:



text



image



multimodality

This success has fostered a dominant narrative, by Anderson (2008):

— “This is a world where massive amounts of data and applied mathematics replace every other tool that might be brought to bear. [...] With enough data, the numbers speak for themselves.”

Introduction: Data-Driven vs. Structural Models

Is Big Data + Black Box all we need? Are domain-specific theories any useful?

- Big Data + Black Box: solely data-driven → highly expressive, poor generalization
- Domain-specific theories: structured knowledge → interpretable, misspecification

How to reconcile such a dichotomy remains an open question...

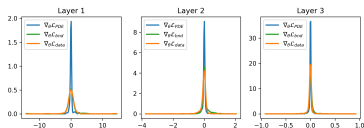
- In engineering, physics, we have PINNs to enforce physical laws (PDEs)
- What about in finance, economics?

Introduction: Brief Literature Review

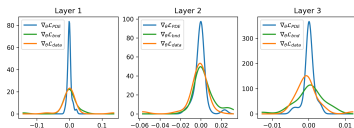
- Raissi et al. (2017, 2019): PINNs which train/penalize NN with PDE residual
↪ Most work solves PDE; little work fuses data- and PDE-knowledge
- Aboussalah et al. (2024); Hainaut and Casas (2024); Wang et al. (2023): further empirical works on option pricing PINNs
- Ackerer et al. (2020); Chataigner et al. (2020); Zhang et al. (2023); Zheng et al. (2021): model-free shape constraints (boundedness, monotonicity, convexity, etc. of the target function, e.g., static no-arbitrage). → Limited economic insights
- Bai (2024); Chen et al. (2023, 2026): deep surrogate modeling; transfer learning
↪ Catastrophic forgetting; heavy (unmanageable) pre-training
- Almeida et al. (2023): correct structural models via learning residuals
↪ Two-stage design; hard to learn from residuals (weak signals)

Introduction: PINNs Limitations

Wang et al. (2021, 2022): Pathological gradient dynamics:



PINN loss gradients w.r.t. the NN weights



PINN loss gradients w.r.t. the NN biases

Unobservable differentiation feature:

$$\partial_T V + \frac{1}{2} v S^2 \partial_{SS} V + \rho \sigma v S \boxed{\partial_{Sv} V} + \frac{1}{2} \sigma^2 v \boxed{\partial_{vv} V} + r S \partial_S V + [\kappa (\theta - v)] \boxed{\partial_v V} - r V = 0$$

PDEs are limited: In finance, economics, domain-specific theories can be prescribed by

- Symbolic equations, e.g., asset/derivative pricing models, time-series models
- Distributions, e.g., $p_t = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[P_T | \mathcal{F}_t]$
- Optimization objective, e.g., portfolio utility

Introduction: Our Contribution

We develop **Structured-Knowledge-Informed Neural Networks** (SKINNs), to better reconcile the dichotomy, particularly for finance, economics:

- SKINNs integrate data learning with structural parameters identification
- Diverse formats of SK to embed domain-specific theories
- Scalable to complex, (ultra-) high-dimensional SK

We demonstrate the power of SKINNs in various finance applications:

- Option pricing → (10%-15% better) OOS pricing & hedging performance
- Asset pricing → (29%-42% better) OOS Sharpe ratio, implementable portfolios
- Realized vol forecasting → (~20% better) OOS forecasting performance

Model: The SKINNs Framework

Consider a data-driven function approximator (NN component):

$$\begin{aligned} f(\mathbf{X} \subseteq \mathbb{R}^d; \theta) : \mathbb{R}^d &\mapsto \mathbb{R}, \\ f^{(l)} &= h\left(\mathbf{b}^{(l-1)} + \mathbf{W}^{(l-1)} f^{(l-1)}\right), \forall l \in \mathbb{N}^+, 1 \leq l \leq L, \\ f^{(0)} &= \mathbf{X} \end{aligned}$$

One typically learn the NN parameters $\theta = \{\mathbf{b}^{(l)}, \mathbf{W}^{(l)}\}_{l=1}^L$ by minimizing a data loss:

$$\mathcal{L}_{\text{Data}}(\theta; \mathcal{D}_{\text{Obs}} \subseteq \{\mathbf{X}_{\text{Obs}}, \mathbf{y}_{\text{Obs}}\}) = \frac{1}{N_{\text{Obs}}} \|\mathbf{f}_{\theta}(\mathbf{X}_{\text{Obs}}) - \mathbf{y}_{\text{Obs}}\|_2^2$$

- **No** awareness of principled scientific structures (no structured knowledge)!
- Result in a local optimal with spurious patterns $\rightarrow$ can't generalize well

Model: The SKINNs Framework

SKINNs embed structured knowledge into the above learning process (SK component):

Definition (Structured Knowledge Representation)

Structured Knowledge is a representation function

$$g(\mathbf{X}^{SK} \subseteq \mathbb{R}^{d_{SK}}; \phi \subseteq \mathbb{R}^{d_\phi}) : \mathbb{R}^{d_{SK}} \mapsto \mathbb{R},$$

which is C^1 -smooth w.r.t. $\mathbf{X}^{SK}, \phi$, or Lipschitz continuous at least, such that ϕ is Gradient Descent learnable. Moreover, $\mathbf{X}^{SK} \subseteq \mathbf{X}$, $d_{SK} \ll d$, $d_\phi \ll d_\theta$.

- The function space of $g_\phi(\mathbf{X}^{SK})$ is given by theories $\rightarrow$ known SK representation
- Its structural parameters $\phi \in \Phi$ are unknown $\rightarrow$ undetermined SK parameters

Model: The SKINNs Framework

Force the data-driven function approximator $f_\theta(\mathbf{X})$ to be SK aware (via a SK loss):

$$\mathcal{L}_{\text{SK}}(\theta, \phi; \mathbf{X}_{\text{Colloc}}) \equiv \frac{1}{N_{\text{Colloc}}} \|f_\theta(\mathbf{X}_{\text{Colloc}}) - \mathbf{g}_\phi(\mathbf{X}_{\text{Colloc}}^{\text{SK}})\|_2^2$$

SKINNs inform the gradient dynamics of training $f_\theta(\mathbf{X})$ through a composite loss:

$$\mathcal{L}_{\text{Total}}(\theta, \phi; \mathbf{X}_{\text{Obs}} \cup \mathbf{X}_{\text{Colloc}}^{\text{SK}}, \mathbf{y}_{\text{Obs}}) = \underbrace{\lambda_{\text{Data}} \mathcal{L}_{\text{Data}}(\theta; \mathbf{X}_{\text{Obs}}, \mathbf{y}_{\text{Obs}})}_{\text{Data fidelity}} + \underbrace{\lambda_{\text{SK}} \mathcal{L}_{\text{SK}}(\theta, \phi; \mathbf{X}_{\text{Colloc}}^{\text{SK}})}_{\text{Theory consistency}}$$

We find the optimal θ, ϕ simultaneously by solving the optimization with (S)GD:

$$(\theta^*, \phi^*) =: \arg \min_{\theta \in \Theta, \phi \in \Phi} \mathcal{L}_{\text{Total}}(\theta, \phi; \mathbf{X}_{\text{Obs}} \cup \mathbf{X}_{\text{Colloc}}^{\text{SK}}, \mathbf{y}_{\text{Obs}})$$

Model: The SKINNs Framework

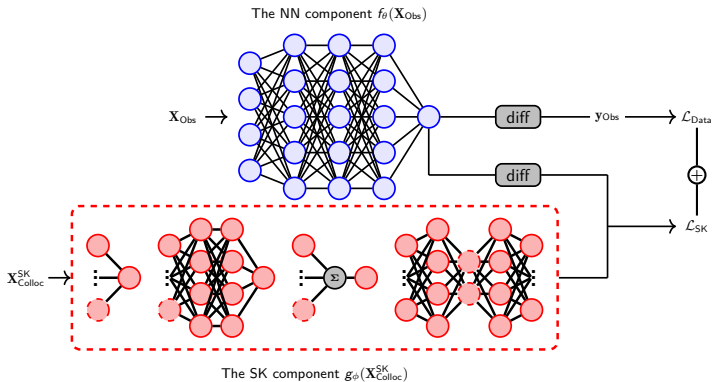
SKINNs work as a generalizability-improved **predictor** & an econometric **estimator**:

- A “white box” to govern a “black box”
- The nuisance θ and the structural ϕ are jointly identifiable (proved in our paper, Section 2.4)
- The estimator-side of SKINNs presents reduced estimation variance than NLS

Our SK representation allows various forms of “theories”:

- Symbolic equations: learn equation parameters
 $\hookrightarrow$ Example: Black-Scholes-Merton model (option), ARMA (realized vol)
- Distributions: learn probabilities
 $\hookrightarrow$ Example: SDF, martingale pricing law
- Downstream optimization objectives: learn optimal decision variables
 $\hookrightarrow$ Example: mean-variance utility maximization

Model: The SKINNs Framework



Model: SK Representation Examples

- Black-Scholes-Merton as the SK rep:

$$\begin{aligned}
 g^{\text{BSM}}(\mathbf{X}^{\text{SK}}; \phi \in \{\sigma\}) &= S_t \Phi(d_1(\sigma)) - K e^{-r(T-t)} \Phi(d_2(\sigma)), \\
 d_1(\phi \in \{\sigma\}) &= \frac{1}{\sigma \sqrt{T-t}} \left(\log(S_t/K) + \left(r + \frac{\sigma^2}{2}\right)(T-t) \right), \\
 d_2(\phi \in \{\sigma\}) &= d_1(\sigma) - \sigma \sqrt{(T-t)}
 \end{aligned}$$

- ARMA(1, 1) as the SK rep:

$$\begin{aligned}
 y_t &= c + \alpha y_{t-1} + \varepsilon_t + \beta \varepsilon_{t-1}, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \\
 g^{\text{ARMA}}(\mathbf{X}^{\text{SK}} = y_{t-1}; \phi \in \{c, \alpha, \beta\}) &= \mathbb{E}[y_t | \mathcal{F}_t] = c + \alpha y_{t-1} + \beta \varepsilon_{t-1}
 \end{aligned}$$

- Unknown Distribution as the SK rep:

$$\begin{aligned}
 g_\phi(\mathbf{X}^{\text{SK}}; \phi \in \{\mathbb{Q}(z^{[i]})\}_i^\infty) &= e^{-r(T-t)} \mathbb{E}^\mathbb{Q} \left[\underbrace{u(z, \mathbf{X}^{\text{SK}})}_{\text{Payoff function}} \mid \mathcal{F}_t \right] \\
 &= e^{-r(T-t)} \sum_{i=1}^{\infty} u(z^{[i]}, \mathbf{X}^{[i], \text{SK}}) \mathbb{Q}(z^{[i]})
 \end{aligned}$$

Model: SK Representation Examples

- Downstream Mean-Variance Optimization as the SK rep:

$$\begin{aligned} \mathbf{w}^* &:= \arg \min_{\mathbf{w}} - \mathbf{w}^\top \hat{\mathbf{r}} + \eta \mathbf{w}^\top \Sigma \mathbf{w} \\ \text{s.t. } & \mathbf{w}^\top \mathbf{1} = 1 \\ & \mathbf{l} \leq \mathbf{w} \leq \mathbf{u} \end{aligned}$$

The MVO is typically solved given a learned asset-pricing model $\hat{\mathbf{r}} = f_\theta(\mathbf{X})$
 $\hookrightarrow$ Two-stage optimization; no interaction with each other

Turn such a two-stage optimization into an SK component:

$$\begin{aligned} g^{\text{MVO}}(f_\theta, \phi \in \{\mathbf{w}\}) &= \mathbf{w}^\top f_\theta + \eta \mathbf{w}^\top \Sigma \mathbf{w} \\ \text{s.t. } & \left. \begin{aligned} & \mathbf{w}^\top \mathbf{1} = 1 \\ & \mathbf{l} \leq \mathbf{w} \leq \mathbf{u} \end{aligned} \right\} \text{Pass to clamping functions} \end{aligned}$$

Similar concept used in Jensen et al. (2026); Wang et al. (2026)

Empirical Study: Option Pricing

We use comprehensive S&P 500 index option data:

- Daily S&P 500 index call options (European style).
- Covering the period from January 4, 1996, to December 31, 2022.
- Options with maturities between 7 and 365 calendar days.

Model training schedule:

- Train on three-month option panels
- Test over the next two consecutive months (first month: shorter prediction horizon; second month: longer prediction horizon)
- 317 training & test subsamples in total

Empirical Study: Benchmark Models

Model	Structured Knowledge	$\dim(\phi)$ (learnable/total)	Activation function
Benchmark NNs			
NN	No	0/0	ReLU
NN+Bnd	Bnd	0/0	ReLU
NN+Shape	Shape	0/0	SiLU
PINN+BSM	BSM	1/1	SiLU
TLNN+HSV	HSV	0/5	ReLU
(PSKR) SKINNs			
SKINN+BSM	BSM	1/1	ReLU
SKINN+HSV	HSV	5/5	ReLU
SKINN+ABSM	ABSM	6/6	ReLU
SKINN+HSVJ	HSVJ	9/9	ReLU
SKINN+SABR	SABR	722/722	ReLU
(SPSKR) SKINNs			
SKINN+DSNN-HSV	DSNN-HSV	5/5	ReLU
SKINN+DSNN-NASV	DSNN-NASV	6/6	ReLU
(NPSKR) SKINNs			
SKINN+AE-BSM	AE-BSM	2/2	ReLU
SKINN+AE-MIX	AE-MIX	50/50	ReLU
SKINN+MOA	MOA	2,000/2,000	ReLU

Empirical Study: OOS Pricing Error (Diebold-Mariano Test)

Longer prediction horizons (RMSE)								
Model	Structural models				Benchmark NNs			
Panel (A)	BSM	ABSM	HSV	HSVJ	NN	NN +Bnd	NN +Shape	PINN +BSM
BSM	—	2.49***	0.62	-5.23***	2.67***	1.36*	2.51***	2.54***
ABSM	—	—	-0.57	-5.83***	2.56***	1.26	2.40***	2.42***
HSV	—	—	—	-12.95***	2.63***	1.33*	2.45***	2.47***
HSVJ	—	—	—	—	3.09***	1.74**	2.94***	2.97***
NN	—	—	—	—	—	-2.51***	-0.44	-0.38
NN+Bnd	—	—	—	—	—	—	1.30*	1.14
NN+Shape	—	—	—	—	—	—	—	0.01
PINN+BSM	—	—	—	—	—	—	—	—
Model	PSKR				SPSKR		NPSKR	
Panel (B)	SKINN +BSM	SKINN +ABSM	SKINN +HSV	SKINN +HSVJ	SKINN +DSNN-HSV	SKINN +DSNN-NASV	SKINN +MOPA	SKINN +AE-BSM
BSM	-3.61***	-3.61***	-3.06***	-3.56***	-2.09**	-2.78***	-2.53***	0.04
ABSM	-3.85***	-3.87***	-3.30***	-3.80***	-2.31**	-3.01***	-2.74***	-0.10
HSV	-3.89***	-3.91***	-3.37***	-3.89***	-2.33**	-3.08***	-2.74***	-0.03
HSVJ	-2.69***	-2.63***	-2.23**	-2.69***	-1.26	-1.88**	-1.76**	0.56
NN	-3.98***	-3.95***	-3.83***	-4.02***	-3.51***	-3.70***	-3.83***	-3.61***
NN+Bnd	-2.72***	-2.69***	-2.63***	-2.77***	-2.28**	-2.44***	-2.61***	-2.54***
NN+Shape	-3.55***	-3.57***	-3.44***	-3.66***	-3.06***	-3.27***	-3.43***	-3.05***
PINN+BSM	-3.50***	-3.53***	-3.42***	-3.62***	-3.02***	-3.23***	-3.41***	-2.82***

Empirical Study: OOS Hedging Error (Diebold-Mariano Test)

Longer prediction horizons (MHE)								
Model	Structural models				Benchmark NNs			
Panel (A)	BSM	ABSM	HSV	HSVJ	NN	NN +Bnd	NN +Shape	PINN +BSM
BSM	—	-4.30***	7.69***	7.67***	3.44***	3.48***	3.52***	3.20***
ABSM	—	—	7.83***	7.81***	3.68***	3.83***	3.72***	3.43***
HSV	—	—	—	-1.42*	0.25	-1.15	1.11	0.20
HSVJ	—	—	—	—	0.39	-0.99	1.21	0.32
NN	—	—	—	—	—	-2.12**	1.40*	-0.05
NN+Bnd	—	—	—	—	—	—	2.76***	1.47*
NN+Shape	—	—	—	—	—	—	—	-1.94**
PINN+BSM	—	—	—	—	—	—	—	—
Model	PSKR				SPSKR		NPSKR	
Panel (B)	SKINN +BSM	SKINN +ABSM	SKINN +HSV	SKINN +HSVJ	SKINN +DSNN-HSV	SKINN +DSNN-NASV	SKINN +MOPA	SKINN +AE-BSM
BSM	-5.07***	-6.33***	0.84	0.35	0.72	1.45*	0.26	0.68
ABSM	-3.66***	-4.96***	1.78**	1.55*	1.45*	2.01**	1.25	1.35*
HSV	-8.32***	-8.12***	-8.42***	-8.11***	-6.37***	-5.63***	-6.69***	-5.64***
HSVJ	-8.26***	-8.03***	-8.34***	-8.07***	-6.18***	-5.58***	-6.47***	-5.57***
NN	-4.71***	-4.59***	-3.44***	-3.63***	-3.41***	-2.89***	-3.58***	-3.83***
NN+Bnd	-5.07***	-5.12***	-3.40***	-3.71***	-3.14***	-2.56***	-3.50***	-4.07***
NN+Shape	-4.41***	-4.46***	-3.49***	-3.67***	-3.31***	-3.03***	-3.60***	-3.88***
PINN+BSM	-4.32***	-4.29***	-3.01***	-3.23***	-2.94***	-2.45***	-3.29***	-3.34***

Empirical Study: SKINNs Outperformance & Market Vol

How is the model's outperformance relative to NN associated with market volatility?

- A bad model performs worse when market vol is increasing (pro-cyclical)
- A good model performs better when market vol is increasing (counter-cyclical)

Check this with: $\Delta \text{RMSE}^{\text{Model}} = \beta_0 + \beta_1 \text{AvgVIX} + \varepsilon$

- $\Delta \text{RMSE}^{\text{Model}}(t_j) := \text{RMSE}^{\text{Model}}(t_j) - \text{RMSE}^{\text{NN}}(t_j)$, $\text{AvgVIX}(t_j) := \frac{1}{N_{t_j}} \sum_{i=1}^{N_{t_j}} \frac{\text{VIX}_i}{100}$
- t_j denotes the OOS evaluation period j , $1 \leq j \leq 317$

Empirical Study: SKINNs Outperformance & Market Vol

Longer prediction horizons (64 low-vol periods)								
Model Panel (A)	BSM	ABSM	HSV	HSVJ	NN +Bnd	–	–	–
β_1	-0.1979 (0.1657)	-0.1172 (0.1644)	-0.1354 (0.1646)	-0.1288 (0.1671)	-0.0346 (0.1192)	–	–	–
β_0	0.0230 (0.0205)	0.0125 (0.0204)	0.0167 (0.0204)	0.0119 (0.0207)	0.0024 (0.0148)	–	–	–
Obs	64	64	64	64	64	–	–	–
Adj. R ²	0.0067	-0.0079	-0.0052	-0.0065	-0.0148	–	–	–
Model Panel (B)	SKINN +BSM	SKINN +ABSM	SKINN +HSV	SKINN +HSVJ	SKINN +DSNN-HSV	SKINN +DSNN-NASV	SKINN +MOPA	SKINN +AE-BSM
β_1	-0.2303 (0.1504)	-0.1690 (0.1528)	-0.0628 (0.1614)	0.0021 (0.1836)	-0.6827*** (0.1828)	-0.1875 (0.1524)	-0.1741 (0.1270)	-0.1343 (0.1275)
β_0	0.0252 (0.0186)	0.0162 (0.0189)	0.0037 (0.0200)	-0.0035 (0.0227)	0.0912*** (0.0226)	0.0221 (0.0189)	0.0175 (0.0157)	0.0167 (0.0158)
Obs	64	64	64	64	64	64	64	64
Adj. R ²	0.0209	0.0035	-0.0137	-0.0161	0.1705	0.0081	0.0138	0.0018

Empirical Study: SKINNs Outperformance & Market Vol

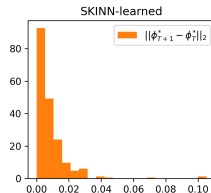
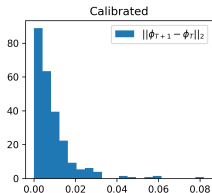
Longer prediction horizons (64 high-vol periods)								
Model Panel (A)	BSM	ABSM	HSV	HSVJ	NN +Bnd	-	-	-
AvgVIX	-0.0364 (0.0783)	-0.0416 (0.0787)	-0.0345 (0.0788)	-0.0272 (0.0791)	-0.0466 (0.0287)	-	-	-
Constant	0.0051 (0.0262)	0.0077 (0.0264)	0.0025 (0.0264)	-0.0013 (0.0265)	0.0094 (0.0096)	-	-	-
Obs	64	64	64	64	64	-	-	-
Adj. R ²	-0.0126	-0.0116	-0.0130	-0.0142	0.0255	-	-	-
Model Panel (B)	SKINN +BSM	SKINN +ABSM	SKINN +HSV	SKINN +HSVJ	SKINN +DSNN-HSV	SKINN +DSNN-NASV	SKINN +MOPA	SKINN +AE-BSM
AvgVIX	-0.1464* (0.0780)	-0.1372* (0.0761)	-0.1516** (0.0739)	-0.1474** (0.0724)	-0.1555** (0.0720)	-0.1373* (0.0767)	-0.1526** (0.0660)	-0.1825*** (0.0453)
Constant	0.0282 (0.0261)	0.0270 (0.0255)	0.0308 (0.0248)	0.0295 (0.0243)	0.0307 (0.0241)	0.0263 (0.0257)	0.0315 (0.0221)	0.0478*** (0.0152)
Obs	64	64	64	64	64	64	64	64
Adj. R ²	0.0385	0.0344	0.0484	0.0475	0.0549	0.0337	0.0645	0.1946

Empirical Study: The Estimator Side of SKINNs

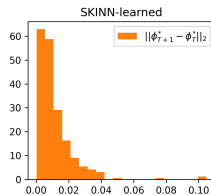
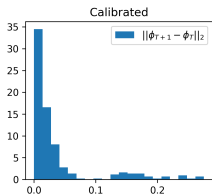
Beyond functioning as predictive devices, SKINNs also work as econometric estimators:

- Nest functional GMM
- Close to semi-parametric models, but more flexible
- (Ultra-) high-dimensional SK parameters are identifiable, with reduced variance
- SKINNs-learned SK parameters carry significant economic information

Empirical Study: The Estimator Side of SKINNs

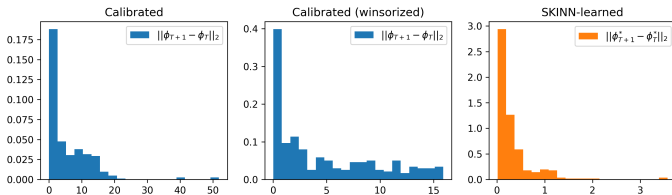


SKINN+BSM

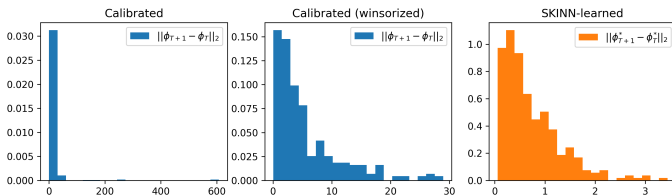


SKINN+ABSM

Empirical Study: The Estimator Side of SKINNs



SKINN+HSV



SKINN+HSVJ

Empirical Study: Asset Pricing

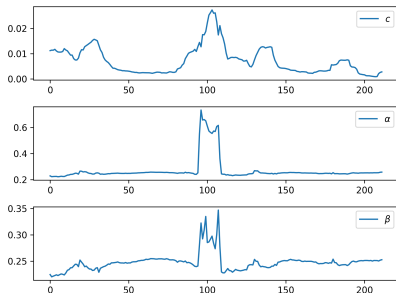
- We trade 50 stocks from S&P 500
- Forward rolling-window test: learn $\hat{r} = f_{\theta}(\mathbf{X})$ using one-year daily OHLC, trade daily over the next month. Stock data spans 2010-2023
- Set $\mathbf{l} = 0.0, \mathbf{u} = 0.2$. Decile-sorting on NN/SKINN predicted returns

Model	Group	NN			SKINN ($\eta = 0$)			SKINN ($\eta = 10$)		
		Avg %	SD %	SR	Avg %	SD %	SR	Avg %	SD %	SR
1-32	H	60.73	130.07	0.47	68.22↑	109.78↓	0.62↑	70.83↑	102.91↓	0.69↑
	L	59.29	136.54	0.43	54.78	133.41	0.41	57.77	134.48	0.43
2-32	H	57.01	127.73	0.45	71.86↑	119.78↓	0.60↑	65.32↑	99.71↓	0.66↑
	L	72.12	145.66	0.50	70.98	141.31	0.50	57.19	140.62	0.41
3-32	H	56.57	127.98	0.44	68.94↑	122.60↓	0.56↑	59.45↑	100.10↓	0.59↑
	L	76.98	132.41	0.58	71.67	132.81	0.54	60.34	134.04	0.45
4-32	H	57.71	124.46	0.46	67.11↑	118.09↓	0.57↑	61.96↑	97.56↓	0.64↑
	L	65.51	136.01	0.48	68.98	133.11	0.52	54.35	134.98	0.40
5-32	H	56.54	125.43	0.45	70.35↑	122.43↓	0.57↑	65.48↑	100.00↓	0.65↑
	L	77.84	134.57	0.58	73.08	132.93	0.55	70.90	135.16	0.52

Empirical Study: Realized Vol Forecasting

- Market (S&P 500 index) 5-min realized vol from January 3, 2000 to 26 June, 2018
- Train on one-year realized vols, test over the next month (212 training & test subsamples in total)
- Features: lagged FF5 factors (daily)

Average OOS RMSE		
Model	NN	SKINN+ARMA(1,1)
	0.0150	0.0121
Diebold-Mariano Test		
NN	—	-2.13**
SKINN+ARMA(1,1)	—	—
Δ RMSE and Market Vol		
β_1	—	-1.8469**
β_0	—	-0.0001
Obs	—	212
Adj. R^2	—	0.0186



SKINN-learned ARMA(1,1) Parameters

Conclusions

- All models/theories are wrong? Feed them to neural networks!
- SKINNs provide a novel unified framework to embed theoretical structures into machines (much more versatile than PINNs, Transfer Learning, etc.)
- Data-driven is not almighty, still need theories for interpretability & generalizability



SSRN